Chapters 8 & 9 (Condensed): Knowledge and Beliefs

ECON 5001
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Based on: *Game Theory* by Giacomo Bonanno

Updated 2026-11-09 at 21:38:59

Where We're Going

So far: everyone knows the game

- Payoffs, strategies, who moves when
- And everyone knows that everyone knows it, and…

Coming up (Chapter 14): players **unsure** about the game itself

- Is my rival's cost high or low?
- Is the entrant tough or weak?
- Does my rival know what I know?

We need a picture of **who knows what**, and **who believes what**

Lite version of Chapters 8–9. Just what we need for Chapters 14–15.

MobLab: “Monty Hall”

- MobLab time!
- Three doors. A car behind one, goats behind the other two
- You pick a door
- Monty (who knows where the car is) opens a **different** door with a goat
- Stay or switch?

Feel free to talk! And play it the way you’d actually play it.

States

State w: a complete description of everything relevant

- $W =$ set of all states
- **Event** $E \subseteq W$: a set of states (“it rains tomorrow”)
- **True state**: the one that actually happened

Information set $I_i(w)$: the states Player i can’t tell apart when the truth is w

- The info sets split up W: a **partition**
- Coarse partition: knows little. Fine partition: knows a lot

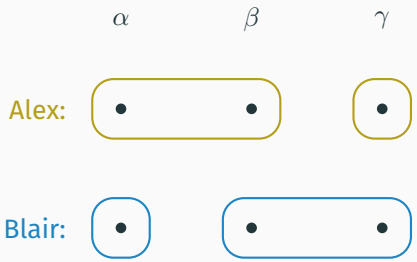
Same word as “information set” in game trees. Same idea: “where could I be?”

Example: Quiz Tomorrow?

Alex was in class. Blair skipped. Alex texts Blair if there's a quiz
Texts sometimes get lost

- α: quiz, Blair got the text
- β: quiz, text got lost
- γ: no quiz (no text)

True state: α

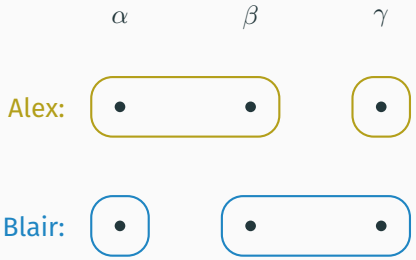


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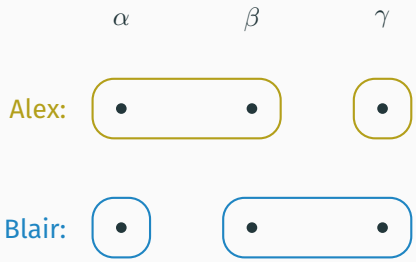
Alex can't tell α from β. Why? _____

Blair can't tell β from γ. Why? _____

Knowledge

In words: *i* knows *E* at *w* if **every** state *i* thinks is possible is in *E*

$$w \in K_i E \iff \underbrace{I_i(w)}_{\text{what } i \text{ thinks is possible}} \subseteq \underbrace{E}_{\text{the event}}$$



$Q =$ "quiz" $= \{\alpha, \beta\}$

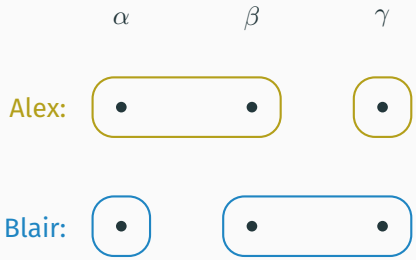
$K_A Q =$ _____

$K_B Q =$ _____

At α, who knows about the quiz?

Knowing What Others Know

"Blair knows Q" is itself an event: $K_B Q = \{\alpha\}$

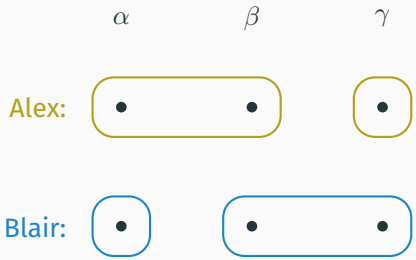


Does Alex know Blair knows?
$K_A K_B Q =$ _____

Does Blair know Alex knows?
$K_B K_A Q =$ _____

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"Blair knows Q" is itself an event: $K_B Q = \{\alpha\}$



Does Alex know Blair knows?
$K_A K_B Q =$ _____

Does Blair know Alex knows?
$K_B K_A Q =$ _____

So at α: both know there's a quiz. Blair knows Alex knows
But Alex **doesn't** know Blair knows!

Common Knowledge

E is **common knowledge** if: everyone knows E, everyone knows everyone knows E, everyone knows everyone knows everyone knows E, ... **forever**

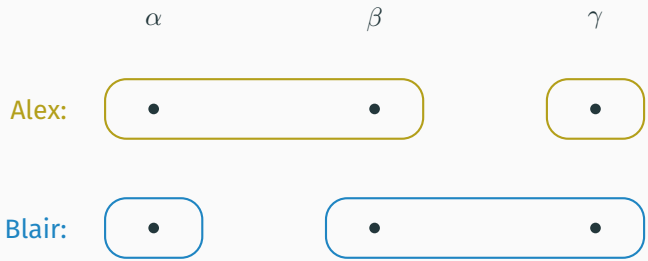
Infinitely many things to check?? No:

Reachable: w' is reachable from w if you can get there in steps, where each step stays inside **someone's** information set

Fact (Bonanno §8.3)
E is common knowledge at $w \iff$ every state reachable from w is in E

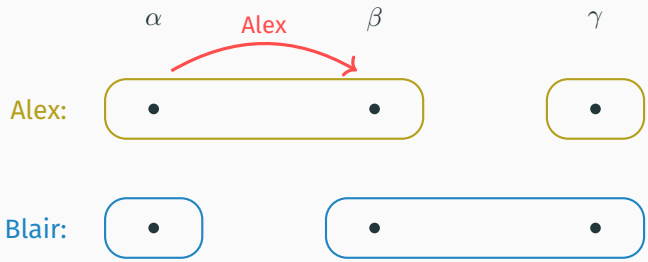
Notation: $CK\,E =$ states where E is common knowledge

Is the Quiz Common Knowledge?



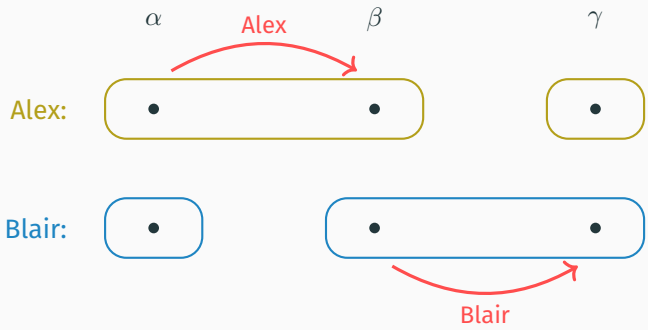
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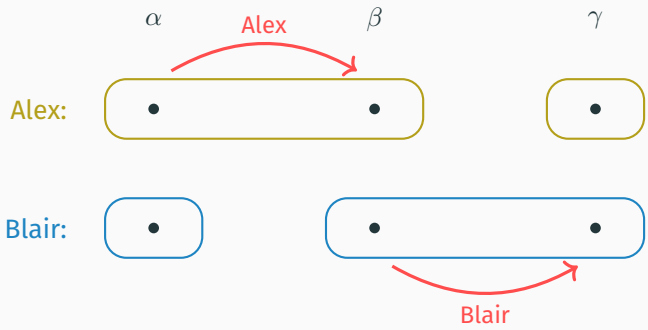
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Starting at α, which states are reachable? _____

$\gamma \notin Q \Rightarrow$ the quiz is **not** common knowledge. Anywhere!

Is the Quiz Common Knowledge?



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$\gamma \notin Q \Rightarrow$ the quiz is **not** common knowledge. Anywhere!

Fact: anything true at **every** state is common knowledge
Ex: "Alex knows whether there's a quiz" **Ex:** the game's rules (Ch. 14!)

Common Knowledge Is Rare, and Powerful

Things that work because they create common knowledge:

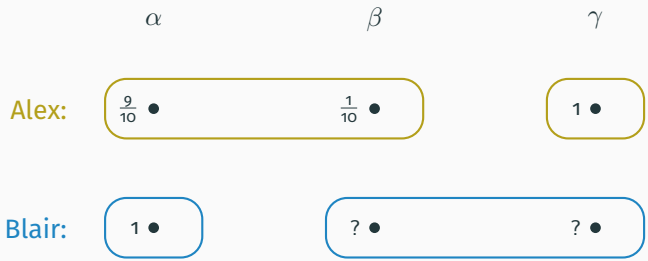
- **Public announcements:** “Not all hats are white” (Bonanno §8.2)
- **Bank runs:** everyone sees the line around the block
- **Super Bowl ads:** you know everyone else saw it too
- **Read receipts:** “Delivered” vs. “Read”
- **Protests:** I’ll go if I know others will go…

Private messages give knowledge. Public events give **common** knowledge

Not in the book. Chwe, *Rational Ritual* (2001).

Adding Beliefs

Each player puts **probabilities** on the states in each of their info sets



Alex: texts arrive 90% of the time

Blair, with no text: Pr(quiz)?

- Need to know: Pr(quiz) at all
- Say Pr(quiz) $= \frac{1}{2}$

Interactive knowledge-belief structure (Bonanno Def. 9.5.1): partitions plus beliefs.

Bayes' Rule

In words: chance of A once you learn B = share of B's probability where A is true

$$P(A \mid B) = \frac{\overbrace{P(A \cap B)}^{A \text{ and } B}}{\underbrace{P(B)}_{\text{all of } B}}$$

Prior: $P(\alpha) =$ _____ $P(\beta) =$ _____ $P(\gamma) =$ _____

Blair gets no text, so learns $\{\beta, \gamma\}$:

$$P(\text{quiz} \mid \text{no text}) = P(\beta \mid \{\beta, \gamma\}) = _____$$

Back to Monty Hall

You pick door 1. A state: (where the car is, which door Monty opens)

State	Prior prob.	Monty opens
(Car 1, opens 2)		2
(Car 1, opens 3)		3
(Car 2, opens 3)		3
(Car 3, opens 2)		2

Monty opens 3. Your info set: _____

$P(\text{car } 1 \mid \text{opens } 3) =$ _____ Switch? _____

MobLab: _____% switched

Common Prior

Common prior: one probability p on W that gives **everyone's** beliefs by Bayes' rule

Ex: Alex at $\{\alpha, \beta\}$: $(\frac{9}{10}, \frac{1}{10})$. Blair at $\{\beta, \gamma\}$: $(\frac{1}{11}, \frac{10}{11})$. Find p:

Bonanno Def. 9.5.2. Existence of a common prior is also called **Harsanyi consistency**.

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Why we care: in Chapter 14, the common prior tells "Nature" how often to pick each state

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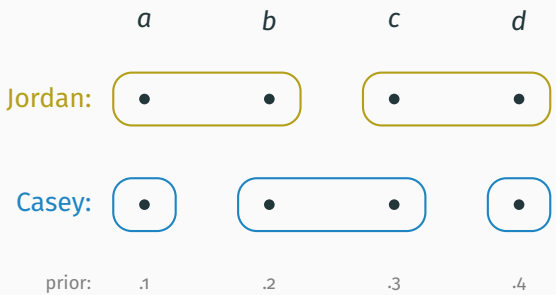
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Fact: a common prior doesn't always exist

Bonanno Def. 9.5.2. Existence of a common prior is also called **Harsanyi consistency**.

Let’s Work One Out

Kickoff Time



Kickoff is at noon in $N = \{a, b, c\}$. True state a

1. Find $K_J N$ and $K_C N$
2. At a, does Jordan know that Casey knows N?
3. Is N common knowledge at a?
4. Use the prior to find each player's beliefs
5. If Casey's beliefs at $\{b, c\}$ were $(\frac{1}{2}, \frac{1}{2})$, is there still a common prior?

What Have We Learned, Charlie Brown?

- **States** and **information partitions**: who can tell what apart
- i knows E at $w \iff I_i(w) \subseteq E$
- **Common knowledge**: every reachable state is in E
  - True at every state $\Rightarrow$ common knowledge
- **Beliefs**: probabilities on each information set
- **Bayes' rule**: $P(A \mid B) = P(A \cap B)/P(B)$
- **Common prior**: one p that generates everyone's beliefs

Reading & Practice

- Bonanno, Chapter 8
  - §8.1–8.3: read. Skip §8.4–8.5 (belief relations)
- Bonanno, Chapter 9
  - §9.1–9.3 and §9.4.1: read (probability, Bayes' rule)
  - §9.5: read (common prior). Skip §9.4.2 and §9.6–9.7
- Exercises: §8.6.1–8.6.3, §9.8.3
- Solutions in §8.7 and §9.9. Try them first